

EDUCATION

Arizona State University

GPA: 3.56 | Aug 2025 – May 2027

M.S. Computer Science · Coursework: Computer Architecture, Statistical Learning Theory, Cloud Computing, Planning and Learning Methods of AI

University of Mumbai

CGPA: 9.04/10 | Dec 2021 – Jun 2025

B.Tech. Computer Science & Engineering(Specialization in IoT, Cybersecurity & Blockchain Technology)

SKILLS

Languages: Python, Go, Rust, C++, Java, JavaScript**AI/ML:** PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Ollama, RAG, MCP**Backend & Infra:** FastAPI, Flask, gRPC/Protobuf, REST APIs, Docker, Kubernetes, RabbitMQ**Databases:** PostgreSQL, Neo4j, MongoDB, Redis, ChromaDB, MySQL**Cloud & Tools:** AWS, GCP, Prometheus, Grafana, Git, GitHub Actions, HuggingFace Spaces

RESEARCH

MOSAIC — Multi-Agent AI Coaching System, ASU VISA Lab

May 2026 – Present

Supervisor: Prof. Ming Zhao | CSCW 2026 (ACM A-ranked, under review)

- Contributing to implementation of a multi-agent AI system integrating mobile behavioural sensing (sleep, social proxy, mobility) with a GROW-framework coaching pipeline across six specialised AgentScope agents.

Malicious URL Classification — MMAUC

Aug 2024 – May 2025

- Ensemble ML pipeline for malicious URL detection; GAN-based augmentation improved class balance 22%, achieving 93% accuracy and 18% fewer false positives. *Under peer review.*

PROJECTS

VulnGraph — AI-Powered ASPM Platform [GitHub]

Dec 2025 – May 2026

Python · Neo4j · ChromaDB · BM25 · llama3.2:3b · Ollama · FastAPI · Streamlit · MCP · HuggingFace Spaces · Render

- Integrated Gitleaks, Trivy, and Bandit scanners into a Neo4j attack graph modelling file-to-vulnerability relationships; deployed on Render with HuggingFace Spaces demo.
- Built a hybrid RAG pipeline (ChromaDB vector search + BM25) over 120 security documents (CWE Top 25, OWASP 2025, Bandit rules) with query expansion and metadata filtering for per-finding retrieval.
- Implemented a deterministic 5-step agentic pipeline (query graph → search KB → explain → read code → patch) producing concrete code fixes; exposed tools as an **MCP server** for Claude Desktop integration.
- Built custom evaluation harness measuring LLM quality on answer relevancy (50%), faithfulness (30%), and CWE accuracy (20%); benchmarked phi3, qwen2.5-1.5B, gemma2-2B, and llama3.2-3B — selected llama3.2 with 67% pass rate (0.600 avg). Tested against dvpwa, secretsandstuff, and OWASP Juice Shop.

Distributed Rate Limiter Service

Nov 2025 – Feb 2026

Go · gRPC · Redis · Kubernetes · Prometheus · Grafana

- Token-bucket rate limiter in Go + gRPC; 3,000 req/s, 1.57ms avg latency, p99 under 12ms across 60,000 requests under sustained load; deployed on Kubernetes with Prometheus/Grafana monitoring and metrics-based autoscaling.

Federated Learning on AWS + Raspberry Pi (IoT Greengrass)

Mar 2026 – May 2026

Python · PyTorch · AWS EC2 · Lambda · S3 · IoT Greengrass · MQTT

- 10-node heterogeneous FL cluster (9 EC2 + Raspberry Pi) with Lambda FedAvg aggregator triggered by S3 events; 97.7% MNIST accuracy in 35.9s. Pi deployed as real IoT Greengrass edge device with MQTT round coordination. Scored 100/100.

Elastic Cloud Application — Custom Autoscaler on AWS

Jan 2026 – Feb 2026

Python · AWS EC2 · SQS · S3 · DynamoDB · Lambda

- Multi-tier face recognition application with custom autoscaling controller (0→15 EC2 instances, **0.11s scale-in**); 0.87s avg latency on 100 concurrent requests; warm cache reduced latency 63%. Scored 100/100.

Self-Healing Distributed AI Training Cluster

Ongoing

Python · gRPC · FastAPI · Docker · PyTorch

- Docker-based master-worker system with heartbeat failure detection, automatic task reassignment, and checkpoint-driven recovery with All-Reduce gradient synchronisation across arbitrary node failures.

PUBLICATIONS

MMAUC URL Classification — Under review · **Blockchain Powered File Sharing System** — Published · **Votix- Decentralized Voting System** — Under Publication

WORK EXPERIENCE

SOC Analyst Intern — Syntel Pvt. Ltd. (Atos/Eviden Group)

Jun–Aug 2023 & Jul–Aug 2024, Mumbai

- Monitored enterprise security alerts using IBM QRadar, ArcSight, LogRhythm, and Zscaler; supported incident escalation and weekly reporting via Power BI with L1/L2 analysts.